

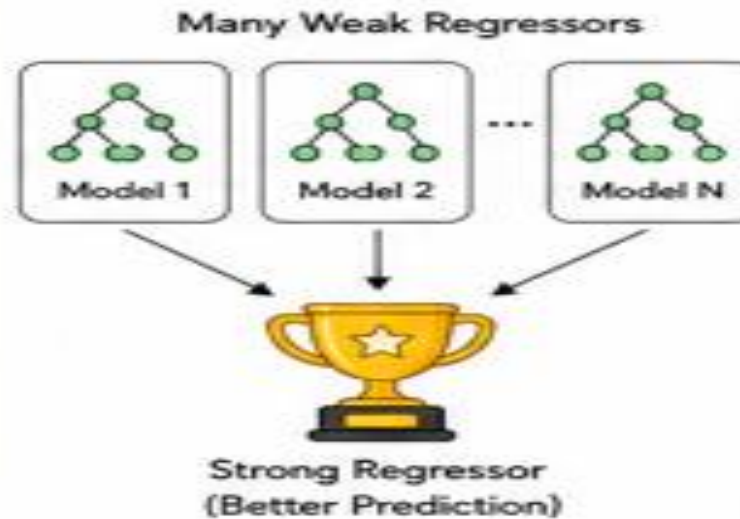
# ADABOOST REGRESSION

## Adaptive Boosting

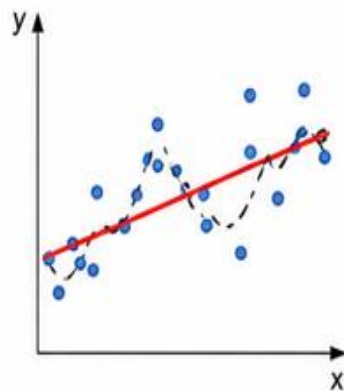
### 1 What is AdaBoost Regressor?

AdaBoost Regressor is a machine learning technique.

- It combines many simple models (regressors) to make a strong model.
- It **learns from mistakes**. More focus is given to difficult cases.



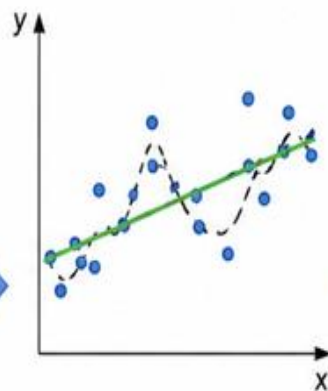
1. Model 1



Model 1 makes the first prediction and has the highest error.

Highest Error

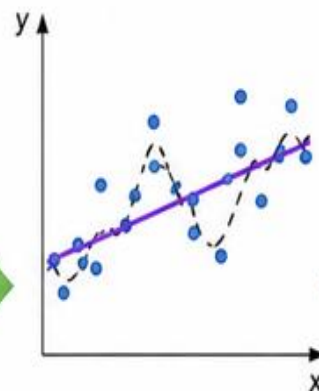
2. Model 2



Model 2 learns from the mistakes of Model 1, so the error decreases.

Lower Error

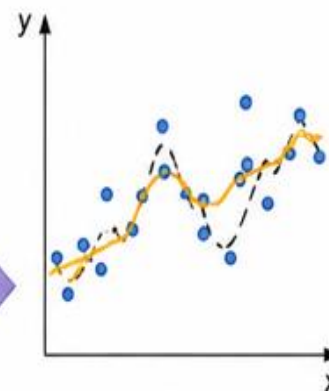
3. Model 3



Model 3 focuses on the remaining difficult data points.

Lower Error

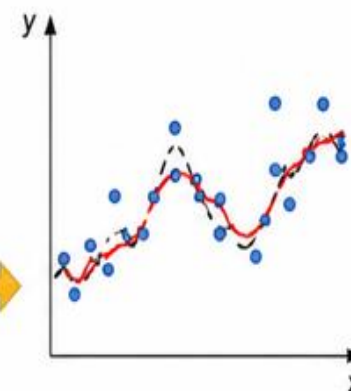
4. Model 4



Model 4 further improves the prediction.

Even Lower Error

5. Final AdaBoost Model



The final AdaBoost model combines all weak regressors to achieve the lowest prediction error.

Lowest Error

--- True function

● Data points

— Model prediction

## 1. Prediction Error

$$e_i = | y_i - \hat{y}_i |$$

- $y_i$  = Actual value
- $\hat{y}_i$  = Predicted value
- $e_i$  = Error

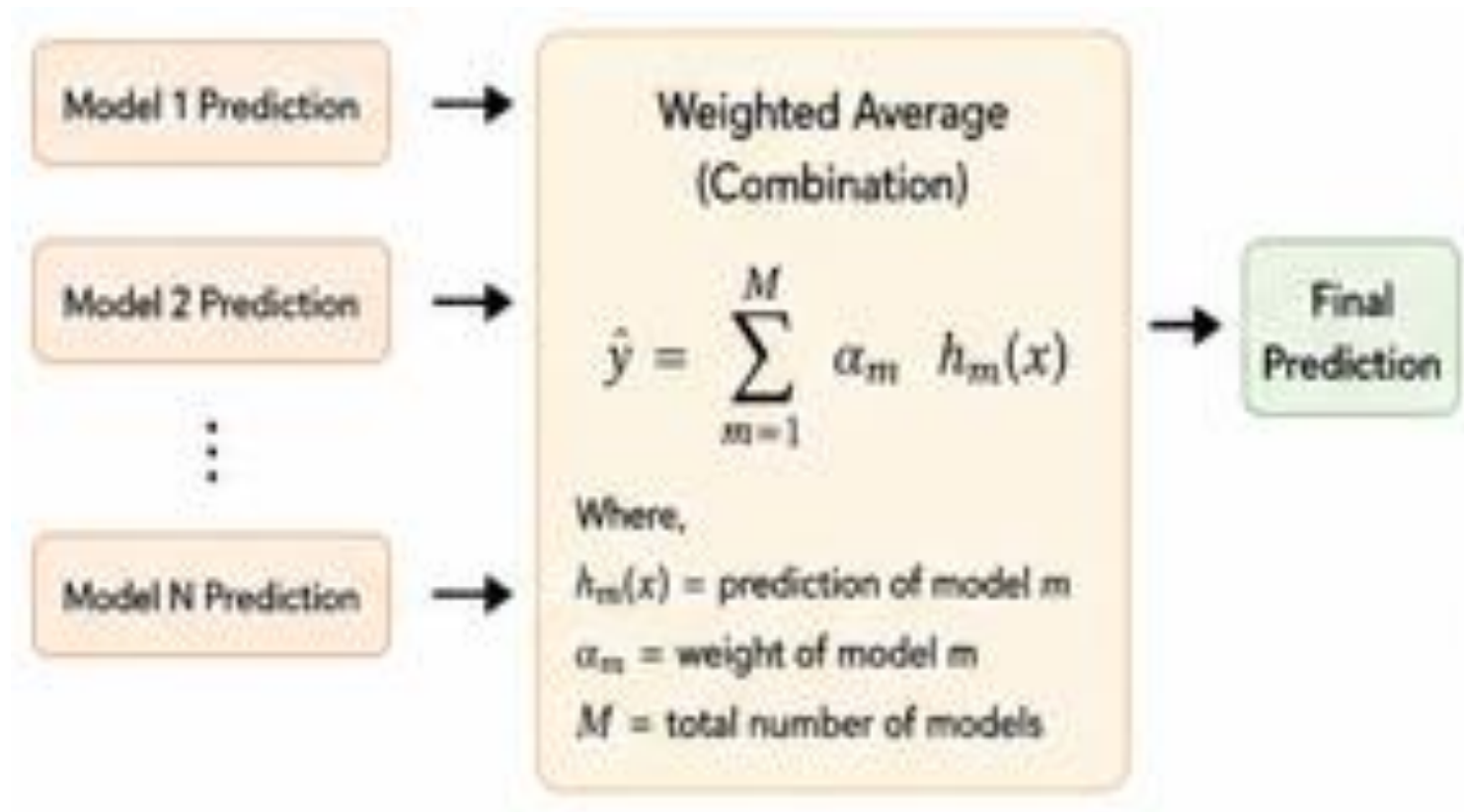
## 2. Final Prediction

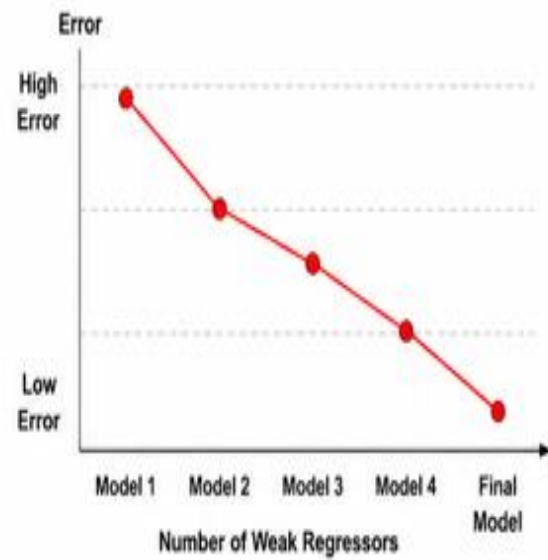
$$F(x) = \sum_{m=1}^M \alpha_m h_m(x)$$

Where:

- $F(x)$  = Final prediction
- $h_m(x)$  = Prediction from the  $m$ -th weak regressor
- $\alpha_m$  = Weight assigned to each regressor
- $M$  = Number of weak regressors

# How predictions are combined?





### Error Progression

Model 1 – Highest Error  
↓  
Model 2 – Lower Error  
↓  
Model 3 – Lower Error  
↓  
Model 4 – Lower Error  
↓  
Final Model – Lowest Error